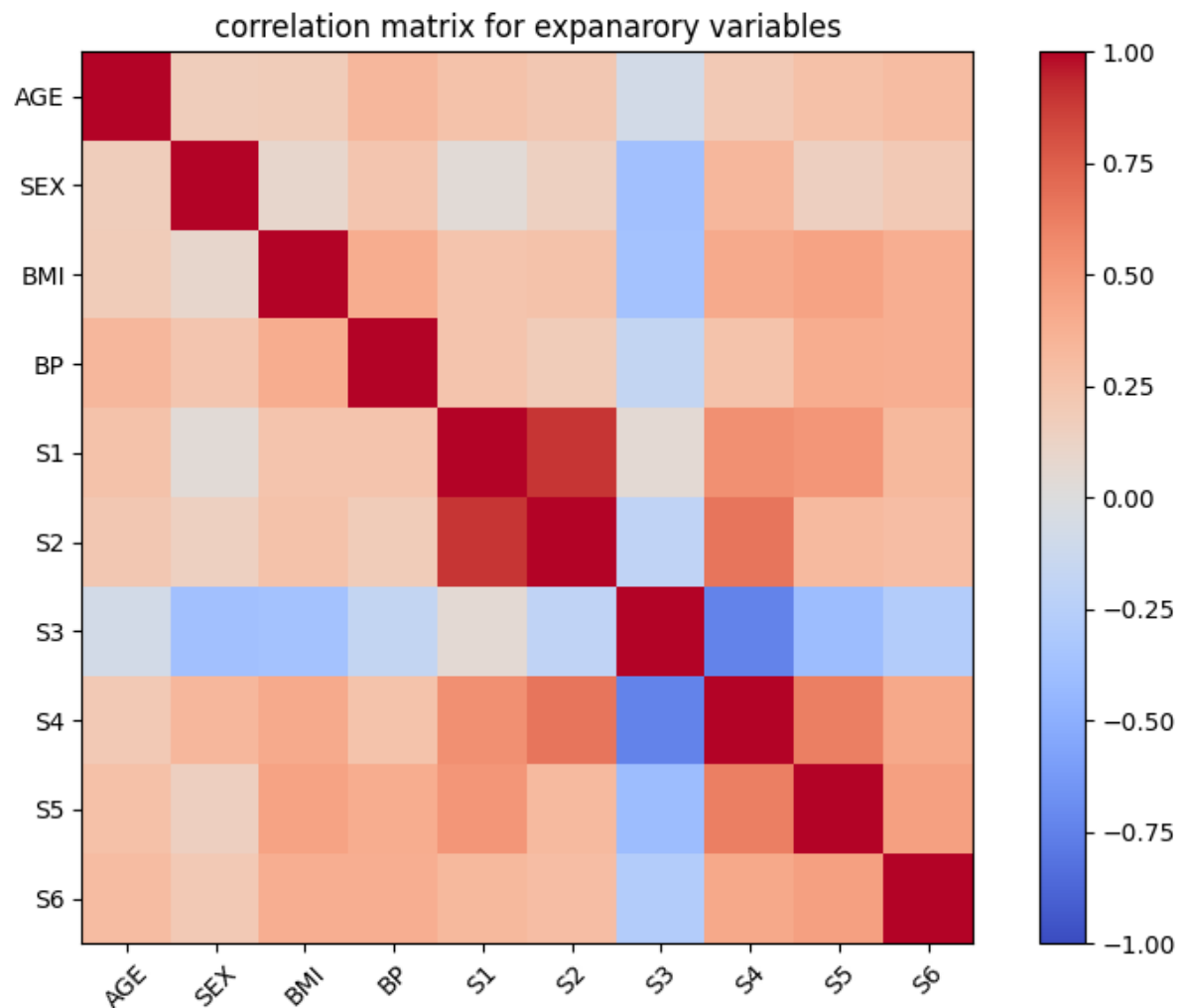


1.1. Relationship between the diabetes dataset explanatory variables according to the correlation matrix heatmap



Colour code description:

- Red - High positive correlation (0.7 - 1)
- faint red - Moderate positive correlation (0.4 - 0.6)
- very faint red - low positive correlation(0.1 - 0.3)
- Blue - High negative correlation (-0.7 - -1)
- faint blue – Moderate negative correlation (-0.4 - -0.6)
- very faint blue - low negative correlation (-0.1- -3)

FOCUSING ON THE HIGH AND MODERATE POSITIVE AND NEGATIVE CORRELATOINS

High positive correlation was observed between
-S1 and S2

Moderate positive correlation was observed between

-S2 and S4

-S4 and S1

-S5 and S4

High negative correlation was observed between

-S4 and S3

Moderate negative correlation was observed between

-S3 and BMI

-S3 and SEX

-S5 and S3

1.2. What is collinearity? What effect does collinearity amongst predictor features/ variables have on their estimated coefficient value?

Collinearity refers to strong linear association among sets of predictors. When occurring among more than one set of variables it is referred to as **Multicollinearity**.

Effect of collinearity

-Increase difficulty in estimation of coefficients

small change in development data can lead to large change in magnitude of estimates and even changes in their signs

-Inflated standard errors

increase in the standard errors of regression coefficients, making it harder to determine whether a variable is statistically significant.

-Difficulty in interpreting the model

challenge in determining individual impact of each variable.

1.3. Are all variables significant

1.4. Difference between forward selection and backward selection

-**Forward selection** involves starting with an empty set of attributes which then is filled with the best attributes after several iterations.

-Model is built **incrementally**

WHILE

-**Backward selection** involves starting with a full of attributes then the worst attributes are removed from the set after several iterations.

-Model is built by **elimination**

1.5. How does the approach stepwise work in the sense of selecting variables?

-The procedure starts with an empty set of attributes as the reduced set.

-The best of the original attributes is determined and added to the reduced set. At each subsequent iteration or step, the best of the remaining original attributes is added to the set.

2.1. What is the difference between logistic regression and linear regression?

-**Linear regression** is a method of predicting continuous values by assuming a linear (straight line) relationship between the input and the output variable.

it is used to estimate quantities such as price, marks, salary, temperature.

-**Logistic regression** is a method used to categorize outcomes into two groups by assuming a Linear relationship between the features and the outcome and then calculating the possibility that the outcome will fall into one group or the other.

it uses the **sigmoid function** which forms a S-shaped curve graph and it converts any number into a value between 0 and 1.

it is used for classification tasks such as a spam mail detector.

3.1. Give a qualitative description of Principal Component Analysis (PCA) and its applications in machine learning. Why might it be useful to consider PCA to transform a set of explanatory variables?

PCA (Principal Component Analysis) is a dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information. It changes complex datasets by transforming correlated features into a smaller set of uncorrelated components

Uses of principal component analysis in machine learning:

-Handles Multicollinearity: Creates uncorrelated variables, making it ideal for datasets with highly correlated features.

-Improves Computational Efficiency: Reduces the number of features, speeding up processing and simplifying models.

-Noise Reduction: Excludes components with low variance, enhancing data clarity

Why might it be useful to consider PCA to transform a set of explanatory variables?

-PCA can be used to reduce the number of explanatory variables under consideration hence improving computational speed, reduce multicollinearity and noise.

3.2.

3.3. similarities between the first and second principal component to the market

-The first principal component closely resembles the market portfolio because it assigns weights of similar sign and magnitude to most stocks.

3.5. Three most distant stocks, why are they unusual.

-The three most distant stocks are **DIS, AXP, and IBM** these have **correlation patterns very different from the average stock.**

Reasons may be

-Highly volatile compared to the rest.

-Low correlation with other stocks